OPERATIONS INTELLIGENCE

Control Tower

YEAR
- [ ] 2024
- [ ] 2025

WAREHOUSE
- [ ] Atlanta DC
- [ ] Chicago DC

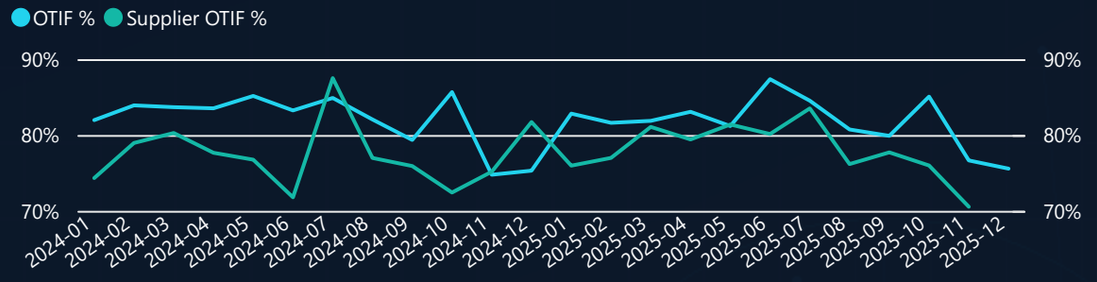
Exceptions first: service, inventory and inbound risk in one operating view.

ORDER OTIF	REVENUE AT RISK	INVENTORY VALUE	STOCKOUT POSITIONS	SUPPLIER OTIF
81.6%	$8.8M	$1.0M	70	77.8%

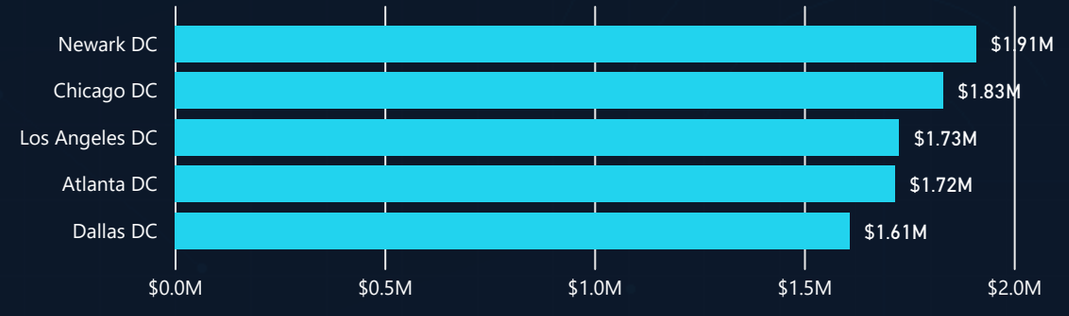
EXCEPTION QUEUE · SKU RISK

SKU	Product	ABCClass	Stockout Risk Locations	Inventory Shortage Value	Revenue at Risk
SKU-0001	Sensors 1001	C	2	$599	$12,324
SKU-0002	Fasteners 1002	C	0	$0	$5,017
SKU-0003	Motors 1003	A	1	$2,594	$608,056
SKU-0004	Pumps 1004	B	1	$342	$26,203
SKU-0005	Sensors 1005	A	2	$3,967	$182,683
SKU-0006	Fasteners 1006	C	1	$83	$10,068
SKU-0007	Motors 1007	C	0	$6	$3,344
SKU-0008	Pumps 1008	C	0	$0	$19,141
SKU-0009	Sensors 1009	A	1	$387	$70,078
SKU-0010	Fasteners 1010	C	0	$92	$24,964
SKU-0011	Motors 1011	A	2	$2,465	$99,698
SKU-0012	Pumps 1012	C	0	$32	$10,810
SKU-0013	Sensors 1013	C	2	$422	$13,389
SKU-0014	Fasteners 1014	C	0	$3	$10,923
SKU-0015	Motors 1015	A	0	$178	$70,606
SKU-0016	Pumps 1016	B	1	$413	$34,200
SKU-0017	Sensors 1017	C	1	$113	$12,123
SKU-0018	Fasteners 1018	A	0	$2,562	$560,809
SKU-0019	Motors 1019	A	1	$1,239	$121,338
SKU-0020	Pumps 1020	C	0	$31	$3,327
Total			**70**	**$53,240**	**$8,786,564**

SERVICE TREND · ORDER OTIF



REVENUE EXPOSURE BY WAREHOUSE



Synthetic operations dataset · 2024–2025

INVENTORY CONTROL

Inventory Intelligence

CATEGORY
- [ ] Accessories
- [ ] Electronics

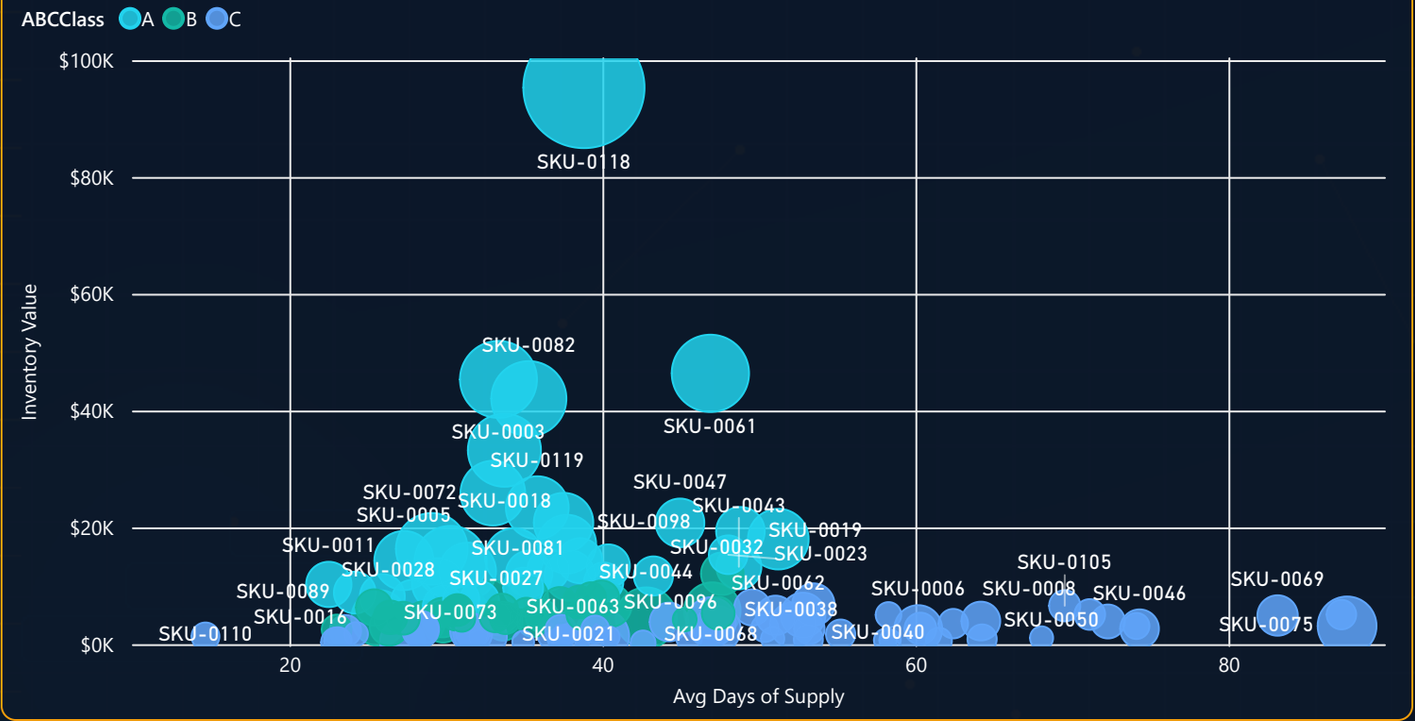
WAREHOUSE
- [ ] Atlanta DC
- [ ] Chicago DC

YEAR
- [ ] 2024
- [ ] 2025

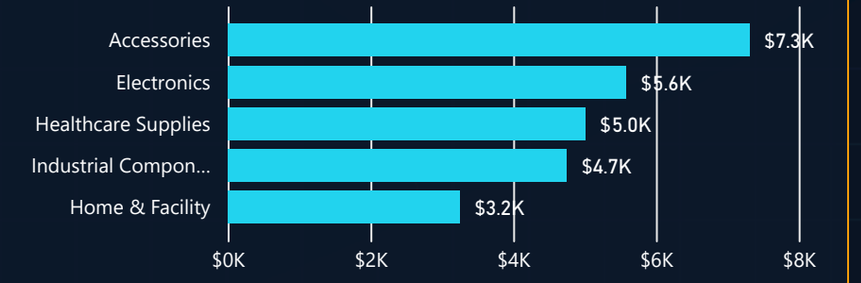
Separate genuine shortages from excess working capital using ABC/XYZ context.

INVENTORY VALUE	EXCESS VALUE	STOCKOUT POSITIONS	AVG DAYS OF SUPPLY
$1.0M	$25.9K	70	41.4

INVENTORY POSITIONING · COVERAGE VS VALUE



EXCESS CAPITAL BY CATEGORY



REPLENISHMENT ACTION QUEUE

SKU	ABCClass	XYZClass	Stockout Risk Locations	Inventory Shortage Value
SKU-0001	C	X	2	$599
SKU-0002	C	X	0	$0
SKU-0003	A	Z	1	$2,594
SKU-0004	B	Z	1	$342
SKU-0005	A	Y	2	$3,967
SKU-0006	C	Y	1	$83
SKU-0007	C	Z	0	$6
SKU-0008	C	Y	0	$0
Total			**70**	**$53,240**

Synthetic operations dataset · 2024–2025

INBOUND PERFORMANCE

Supplier Performance

Prioritize suppliers by service failure, delay exposure and quality—not volume alone.

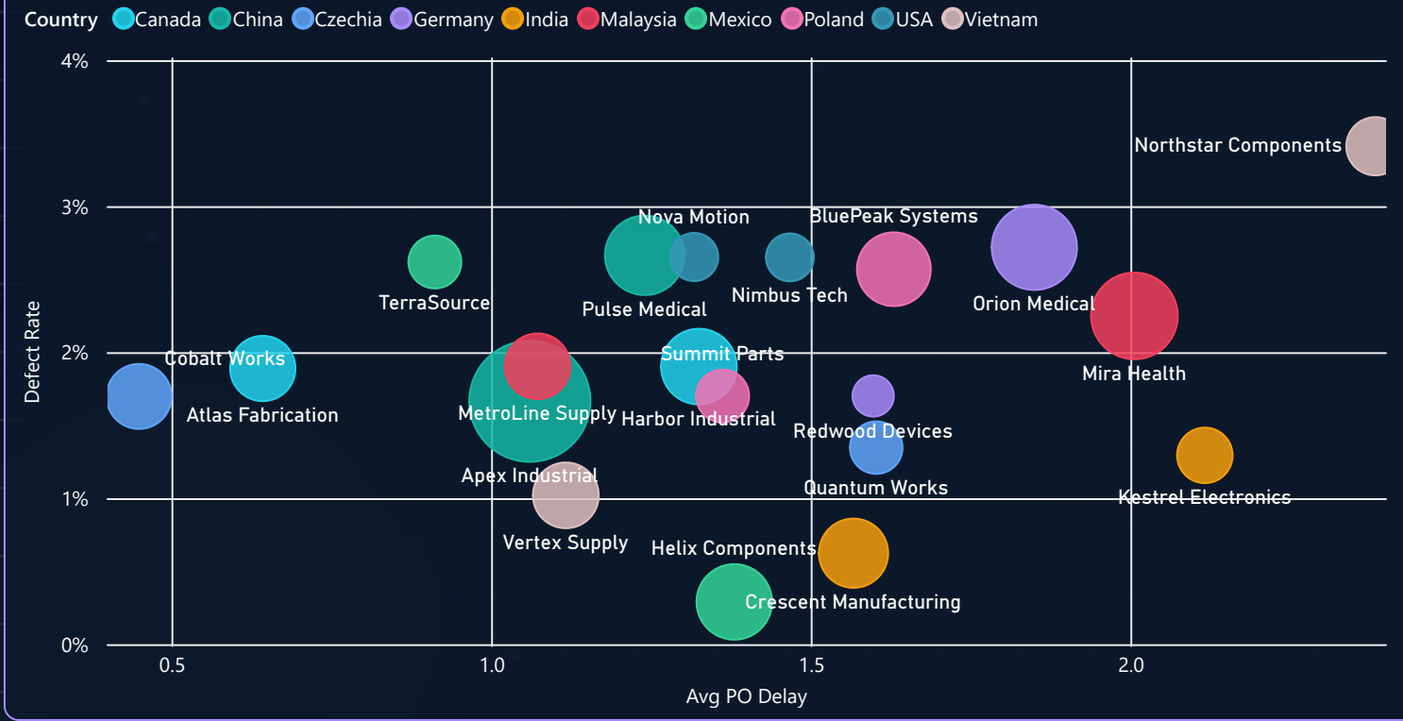
SUPPLIER COUNTRY
- Canada
- China

CATEGORY
- Accessories
- Electronics

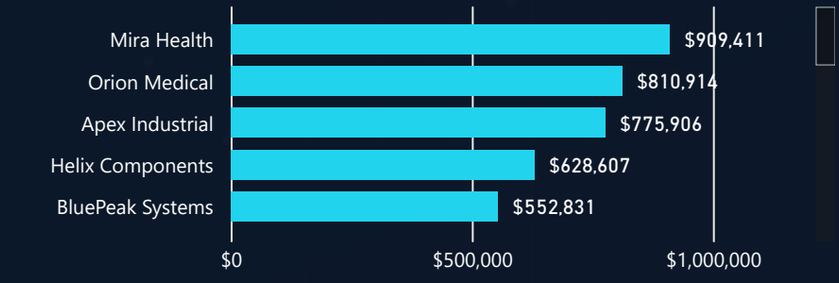
YEAR
- 2024
- 2025

SUPPLIER OTIF	LATE PO VALUE	AVG LEAD TIME · DAYS	DEFECT RATE
77.8%	$7.9M	24.7	1.9%

SUPPLIER RISK MAP · DELAY VS QUALITY



LATE INBOUND VALUE · SUPPLIER



SUPPLIER SCORECARD

Supplier	Country	Supplier OTIF %	Avg Lead Time	Late PO Value
Apex Industrial	China	85.7%	23.8	$775,906
Atlas Fabrication	Canada	90.5%	14.3	$150,610
BluePeak Systems	Poland	74.0%	13.5	$552,831
Cobalt Works	Czechia	94.7%	22.1	$136,763
Crescent Manufacturing	India	75.6%	34.3	$353,134
Harbor Industrial	Canada	76.4%	15.0	$475,524
Helix Components	Mexico	71.7%	14.1	$628,607
Kestrel Electronics	India	63.1%	18.0	$449,643
Total		**77.8%**	**24.7**	**$7,916,635**

Synthetic operations dataset · 2024–2025

CUSTOMER SERVICE

Fulfillment & Logistics

CUSTOMER SEGMENT
- [ ] Enterprise
- [ ] Mid-Market

WAREHOUSE
- [ ] Atlanta DC
- [ ] Chicago DC

YEAR
- [ ] 2024
- [ ] 2025

Connect service failures to customer and revenue exposure, then isolate warehouse pressure.

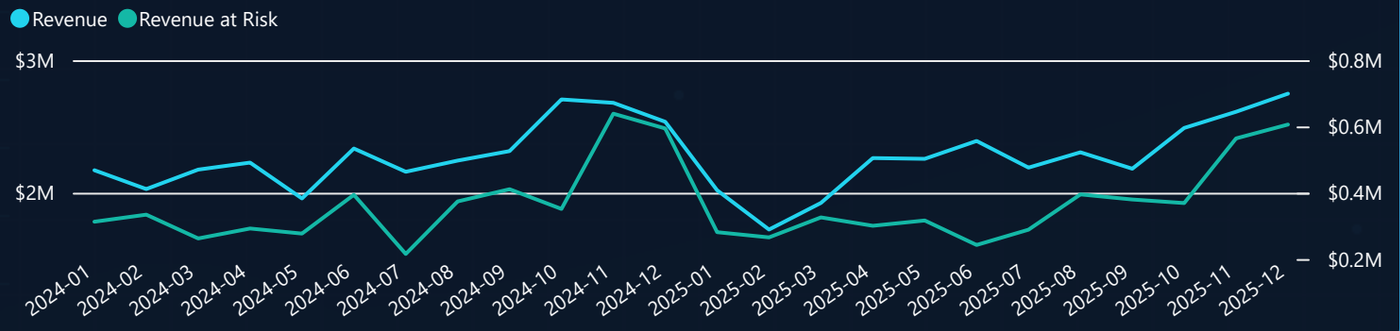
REVENUE
$54.6M

GROSS MARGIN
33.7%

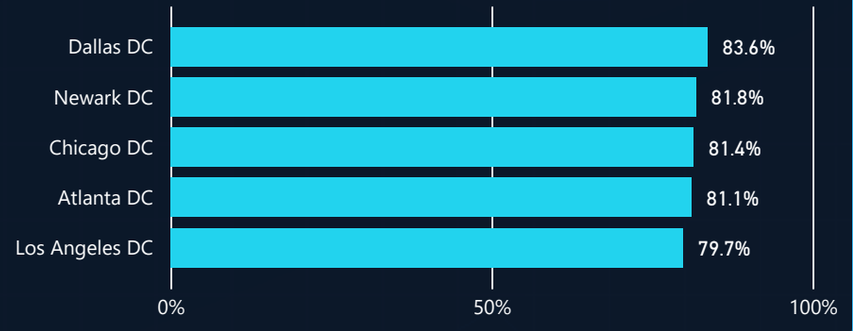
ORDER OTIF
81.6%

REVENUE AT RISK
$8.8M

FULFILLMENT TREND · REVENUE VS RISK



WAREHOUSE SERVICE LEVEL · OTIF



CUSTOMER SERVICE EXPOSURE

Customer	Segment	Industry	Revenue	OTIF %	Revenue at Risk
Customer 001	Retail	Technology	$189,070	80.6%	$28,576
Customer 002	Enterprise	Manufacturing	$264,256	89.0%	$24,650
Customer 003	Mid-Market	Facilities	$195,250	82.3%	$27,307
Customer 004	Retail	Manufacturing	$276,161	87.3%	$32,894
Customer 005	SMB	Public Sector	$217,450	83.6%	$36,589
Customer 006	Mid-Market	Technology	$220,321	84.3%	$29,169
Total			**$54,614,214**	**81.6%**	**$8,786,564**

Synthetic operations dataset · 2024–2025

ENGINEERING PLAYBOOK

Blueprint & Methodology

Report logic, model architecture, refresh setup and portfolio design decisions.

MODEL ARCHITECTURE

Three fact tables anchor the solution: sales orders, purchase orders, and inventory snapshots. Date, Product and Warehouse are conformed dimensions; Supplier and Customer remain process-specific. This creates a clean star-schema experience for drilling from enterprise KPIs into operational detail.

CORE DEFINITIONS

OTIF = orders delivered on-time and in-full. Revenue at Risk = revenue tied to orders that missed OTIF. Stockout position = product/warehouse snapshots below 60% of trailing 28-day demand. Excess inventory = units above 2.5× trailing 28-day demand.

DECISION LOGIC

Control Tower leads with exceptions. Inventory separates service risk from excess capital. Supplier analysis combines delay, quality and value. Fulfillment traces service failures into customer revenue exposure.

DATA & REFRESH

Synthetic but deterministic dataset covering 2024–2025. All CSV sources ship inside /data. After extracting the project, update the DataFolder parameter once, then refresh. This keeps the project portable and Git-friendly.

ENGINEERING CHOICES

Built as PBIP with PBIR and TMDL. Auto date/time disabled. Explicit Date dimension. Single-direction star-schema relationships. Dedicated _Measures table. Hidden technical keys. Measures grouped into Fulfillment, Inventory, Suppliers, Exceptions and KPI display measures.

PORTFOLIO UPGRADE

This version intentionally fixes the numeric-format problem found in the earlier build. Financial measures now preserve detail in tables while KPI cards use dedicated display measures for executive readability. The page system also adds stronger panel treatment and a more deliberate command-center palette.